

BrickAtlas: A Benchmark for Visual Evidence Following with Paired LDraw Observations

Anonymous CVPR submission

Paper ID Draft

D1 | 31028 | 53 parts
Sea Plane



D2 | 42004 | 246 parts
Mini Backhoe Loader



D3 | 10159 | 592 parts
City Airport



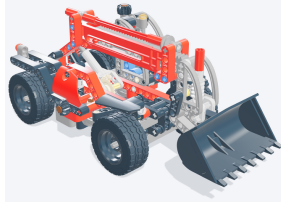
D4 | 10265 | 1,357 parts
Ford Mustang



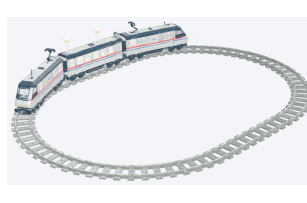
D1 | 42102 | 129 parts
Mini CLAAS XERION



D2 | 42061 | 261 parts
Telehandler



D3 | 10001 | 848 parts
Metroliner



D4 | 42054 | 1,978 parts
CLAAS XERION 5000 TRAC VC

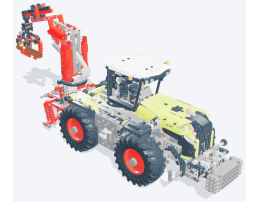


Figure 1. Real source assemblies across D1–D4 scale bands: two examples per band, retaining original LDraw geometry and poses. Source IDs and part counts connect each render to the released catalog. These assemblies provide provenance. Figures 2–4 develop source-backed repair, diagnosis and registration problems. The supplement depicts all 24 sources and the atomic visual controls.

Abstract

An answer can be correct without depending on the visual evidence that the question selects. Evaluating this dependence requires observations with controlled changes, unambiguous references, and a scoring protocol that retains both successful and failed responses. We present BrickAtlas, a benchmark built from 24 human-authored LDraw assemblies containing 15,334 placed parts. Its primary visual track contains 73 rendered-color pairs and 67 candidate-reference pairs; 67 position panels and 219 explicit text-graph observations provide separate diagnostic controls. A display-local reference contract distinguishes the object selected in the current image from its historical source identity. Construction combines exact input records, bounded answer-prior audits, and an independent human qualifica-

tion protocol. Evaluation reports endpoint accuracy, both-correct pair accuracy, conditional adaptation, old-answer retention, and format validity, with source and author dependence made explicit. We provide a study design spanning proprietary and open-weight vision–language models and text-only graph baselines. Three separate compositional case studies demonstrate minimum repair, adaptive fault diagnosis and rigid registration on real subassemblies. The resource and deterministic audits are available; human qualification and the proposed 15-model study remain pending. Experiment tables specify the planned comparisons and leave all unmeasured model results blank.

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1. Introduction

Vision–language models often answer questions about small objects, their appearance, and the references that identify them. Ordinary accuracy measures whether an answer is correct on a supplied image. It does not establish whether the model will remain correct when the relevant evidence changes. A model may exploit answer frequency, option position, a familiar source, or a stable reference convention. A paired benchmark asks a more demanding question: can the model answer both observations correctly under the same task contract?

Constructing such a benchmark introduces a second measurement problem. The intervention must change the intended answer while preserving what the question means. A part’s source identifier can denote an immutable record, whereas a label in an edited image can select a different visible object. Without resolving this distinction, an apparent failure to adapt may reflect an inconsistent question. Similarly, a source record can certify a part type without proving that the distinguishing geometry is visible to a human.

BrickAtlas addresses these problems through an auditable LDraw resource [7]. Source files provide exact geometry, transforms, part types, and authorship. Rendered observations expose only the evidence permitted by the question. Unique numeric labels, such as B0035, identify image-local referents, while source identities remain available to the evaluator. Every endpoint has a fixed question, option list, image, gold answer, and provenance record.

The frozen primary benchmark has two atomic visual families. **Color** is an appearance sanity check about the currently rendered color of a single labeled part. Its paired intervention changes appearance while keeping geometry and pose fixed. **Part-type** asks which labeled candidate matches a target’s visible geometry. The target stays fixed while two candidate labels exchange referents. This tests geometry matching together with selection of the correct current label. We report the two families separately because their visual requirements differ. Beyond these controls, three newly designed examples combine graph intervention, conditional inspection and spatial constraints on complete source sub-assemblies. They illustrate richer task contracts and exact ground truth, while remaining separate from the frozen visual study.

The evaluation connects three questions. First, is the answer well defined in the exact observation? Second, can declared shallow rules exploit construction regularities? Third, after these checks, does paired performance reveal behavior hidden by ordinary accuracy? The first two motivate construction and qualification; the third motivates the planned experiments. A separate text-graph track illustrates how matching low-level statistics can make a control more informative, without treating graph performance as a visual score.

Our contributions are a source-backed paired resource with explicit display semantics; a construction and qualification protocol linking stimuli, references and review records; and an evaluation design combining pair outcomes, legal input controls and clustered uncertainty. Pair accuracy and controlled perturbations are established methods. The intended value here is their auditable application to this specific resource. Learned-model diagnostic value remains an empirical question, rather than a conclusion of the construction process.

2. Related Work

Controlled visual reasoning. CLEVR [5] and GQA [4] connect questions to explicit scene structure. VQA v2 uses similar images with different answers to reduce language-only shortcuts [3]. Contrast sets probe local decision boundaries [2], and cycle-consistency studies answer stability under linguistic variation [10]. BrickAtlas uses source structure to construct and audit inputs while keeping source records outside the primary visual request.

Paired evidence and reference grounding. Winoground evaluates joint image–text compatibility [12]. Its difficulty analysis highlights the need to distinguish intended reasoning from visibility and other requirements [1]. Numeric-label selection is a restricted reference task compared with natural referring expressions [9]. VisualFLIP directly studies independently queried answer-changing image pairs and competence-conditioned failures [14]. Its v1 preprint reports 1,374 images, four perturbation categories and 24 MLLMs. BrickAtlas does not introduce paired correctness or claim broader coverage. Its focus is the contract among source identity, current display reference, restricted assignment audits and exact-input human qualification.

Brick assembly and structural reasoning. BrickNet uses graph-backed assembly programs [6]; LEGO-Puzzles evaluates multistep spatial reasoning [11]; PhyBlock studies physical understanding and planning [8]. BrickAtlas’s current primary endpoint is atomic visual evidence following. Its historical action dossiers and 3D replay support inspection, but do not establish planning or physical execution ability. The supporting graph construction instantiates a known anonymous 1-WL limitation [13], not a new expressivity theorem.

3. Benchmark Definition

3.1. Resource, tracks and units

The source resource contains 24 original assemblies, 15,334 placed parts and 617 historical task dossiers. The paired visual benchmark derives from 140 parents in that resource.

Table 1. Resource units. Visual families are scored separately. Graph change and invariance pairs share anchors. Historical dossiers are not additional primary visual observations.

Component	Parents	Observations
Color	73	146
Part-type	67	134
Position panels (auxiliary)	67	67
Hypothetical text graphs	73	219
Source assemblies		24
Placed source parts		15,334
Historical task dossiers		617

A parent contributes one A/B pair; a different image of the same parent is not a new source. Table 1 separates primary observations, auxiliary observations and historical material.

Both visual families cover all 24 sources. Source bands D1–D4 describe assembly scale, not validated cognitive difficulty. The supplemental source table reports each assembly and its contribution. The historical dossier remains searchable and links to 3D rotation, zoom, isolation, expansion and given-order replay. These richer inspection views are not primary model inputs.

3.2. Compositional case studies

We include three source-backed examples, CX1–CX3, that require multiple constraints to be satisfied jointly. They have public input records, independently checkable GT and original-geometry renders. These three examples are not an evaluated task family or evidence of model difficulty. The supplement gives every question, adjacency, coordinate and verification certificate.

CX1: multi-fault minimum repair. The 36-part backhoe subassembly of OMR 42004 induces 44 recognized connector pairs and nine independent graph cycles. Eight specified vertices become unavailable, splitting the supplied graph into five components. The task asks for every minimum-cost restoration that reconnects four terminal IDs, the final component sizes, and whether a budget of two is sufficient. It combines path reasoning, competing repairs and an optimality claim. Exhaustive evaluation of all 256 subsets yields the unique minimum $\{B0160, B0175, B0177\}$, cost three, and one 31-vertex final component. All 37 subsets with cost at most two fail. Figure 2 contracts only the surviving components; the original graph remains the scored input.

CX2: adaptive fault diagnosis. On the same source graph, four candidate worlds each disable three vertices. Queries ask whether two IDs remain connected. The answer must be a conditional policy that identifies the exact

world with minimum worst-case cost. Figure 3 gives the hypotheses and computed policy: query Q3 first, then choose Q1 or Q2 from its outcome. The worst-case cost is two; no single binary query can distinguish four worlds. Correctness is checked on every world, not on one successful query sequence.

CX3: correspondence and rigid pose recovery. Two actual 12-part axle assemblies in OMR 42061 instantiate the same design at different source transforms. Repeated part types admit four type-compatible anchor assignments. The solver must determine the unique admissible correspondence, fit a proper rigid transform, and transfer a probe’s origin and oriented local axis. Figure 4 includes all eight anchor coordinates and the probe. Only one assignment admits maximum error below .05 mm: $R = \text{diag}(-1, 1, -1)$ and $t = (16, 0, 120)$ mm. All twelve complete source part transforms verify this result. The other assignments violate pairwise-distance bounds, so their rejection does not depend on a particular numerical optimizer.

The examples have no learned-model or human results and are not added to the 140 paired visual tests. Their complexity is specified by operations and constraints, not inferred from source-scale bands. Graph connectivity is an explicit abstraction; the examples make no claim about physical removal paths, clutch forces or stability.

3.3. Source identity and current display reference

Let $c = (h, i)$ identify a source instance by source-file hash h and part ID i . Its record $T(c)$ contains the original type, color and transform. For observation o , the display mapping $d_o(\ell)$ selects the object marked by label ℓ . The answer y_o follows the current question and d_o , rather than necessarily retrieving a property from $T(c)$.

Every visual question explicitly states that references select objects in the supplied image, that appearance or label assignments may be edited, and that the answer must follow the current image. The model returns one JSON object with a listed `choiceId`. Canonical records, intervention meta-data, golds and pair membership remain evaluator-only.

For Color, the question is “What color is the object currently marked B0035 in this image?” For Part-type, it asks which displayed candidate label currently marks an object with the same visible part geometry as the target, ignoring color and pose. A label reassignment changes the latter answer without changing a canonical part’s type. The schema key `shape-match` refers to this Part-type family.

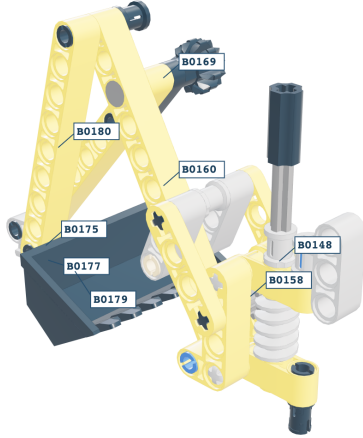
3.4. What each observation permits

A primary input contains the complete system/user text, four options and one native 1280×800 PNG. A and B are independent requests; neither includes the counterpart, its answer or conversational history. External crops, source

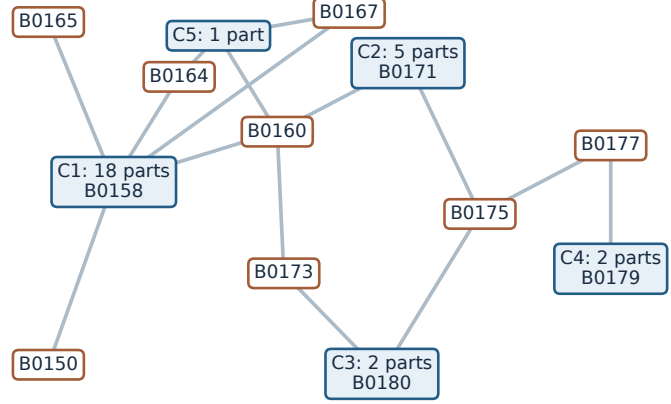
CX1 | Multi-fault minimum repair | OMR 42004

Restore the fewest of 8 missing vertices to reconnect B0158, B0171, B0179 and B0180. Every restore reinstates the original incident edges. Is a budget of 2 sufficient?

Actual 36-part source subassembly



Exact component contraction after faults



Missing: B0150, B0160, B0164, B0165, B0167, B0173, B0175, B0177

GT: {B0160, B0175, B0177}; minimum cost = 3; budget 2 is infeasible.

Initial component sizes: 18, 5, 2, 2, 1. After the minimum repair: one 31-vertex component. All 256 subsets are enumerated; none of the 37 subsets of size 0, 1 or 2 succeeds.

Figure 2. CX1: an original 36-part LDraw subassembly and a constrained minimum-repair problem. Blue boxes summarize surviving components; their terminal IDs and sizes are shown. Brown-bordered vertices are unavailable candidates, with original incident edges retained in the diagram for reasoning about restoration. GT requires a globally minimal set, rather than any successful repair. The native 3D view shows the intact source: faults are abstract graph interventions, not simulated extraction or unsupported physical poses.

CX2 | Adaptive diagnosis under a query budget

Which three-vertex fault world holds? Query path connectivity; minimize the worst-case number of tests.

Possible unavailable sets

W1: B0150, B0160, B0175

W2: B0150, B0173, B0175

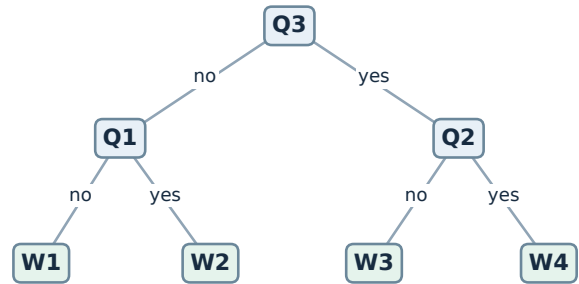
W3: B0150, B0164, B0175

W4: B0150, B0164, B0165

Each test costs 1. All tests start at B0158.

Q1: reach B0171 Q2: reach B0179

Q3: reach B0180



GT: Q3 first; worst-case cost = 2. Budget 1 fails.

Figure 3. CX2: minimum-cost conditional inspection. Each fault world defines a different vertex-unavailability set on CX1's original graph. Reachability outcomes are recomputed from that graph, not supplied as an answer table. The optimal second query depends on the first result. The tree is algorithmic GT; no model policy or physical sensor test has been collected.

CX3 | Typed correspondence + rigid pose recovery | OMR 42061

Match the four anchors by part type and geometry; recover $q = R p + t$ with $\det(R) = +1$.
Maximum error: 0.05 mm. Predict the transformed origin and local x-axis of B0030.

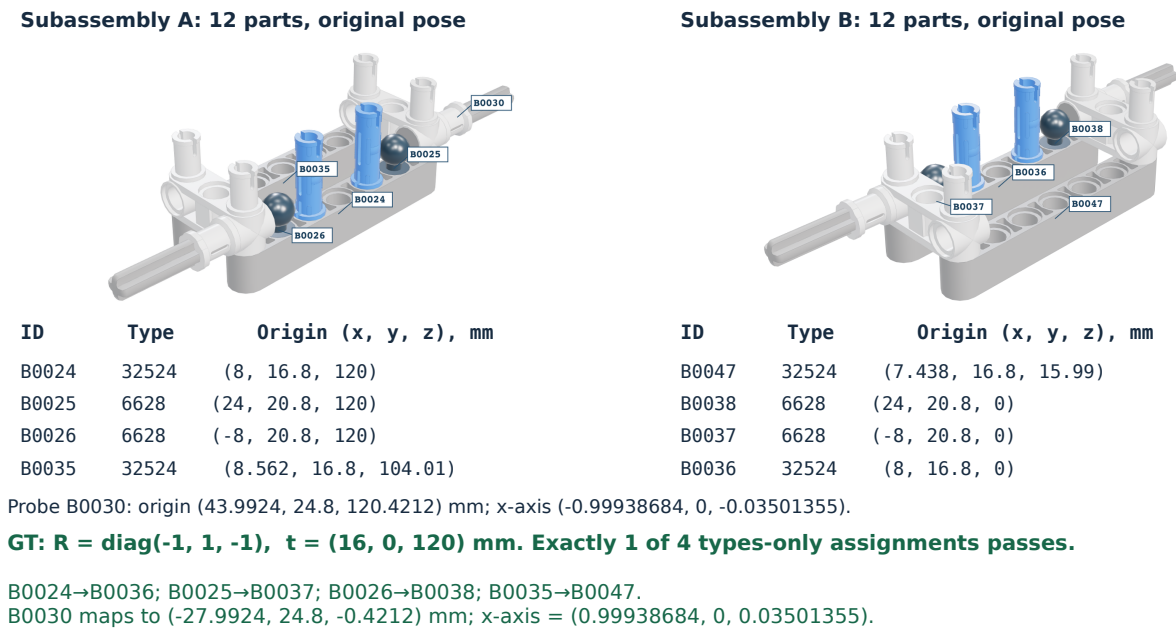


Figure 4. CX3: two complete source axle assemblies, duplicated part types, four-anchor matching, proper rigid registration and probe transfer. Source geometry and all internal poses are preserved. Coordinates specify part origins in world millimeters. GT includes correspondences, rotation, translation, transformed probe location and direction. These source-data constraints go beyond choosing one matching part or identifying an appearance.

records, masks and tools are excluded from the primary visual condition. Provider image policy and generation settings belong to the study configuration.

The source geometry and saved poses remain unchanged. Operand-only views can omit surrounding support and therefore represent observations of parts, not freestanding assemblies. Expansion similarly supports inspection. Recognized connector graphs and preserved source meshes do not certify complete mating, collision-free insertion or stability. The resource retains 573 intersection candidates requiring physical review; the present visual benchmark does not resolve those judgments.

4. Construction and Qualification

4.1. Controlled visual changes

Color. Each of 73 pairs renders one operand and one label. Only the body-color parameter changes between two distinct named opaque colors; geometry, source pose, camera, scale, visibility and label projection stay fixed. Across 146 endpoints the displayed golds are Red, Blue, Yellow and

Black with counts 37, 37, 36 and 36. All pairs have zero changed pixels outside the common projected body mask with one pixel of antialiasing dilation. Three independently regenerated canaries are byte-identical. These are construction checks, not human color judgments.

Color is an appearance sanity track. A legal whole-image color baseline receives only the current PNG and text/options, without a mask, source record or counterpart. Its frozen thresholds yield 141/146 correct endpoints, five abstentions, and 68/73 Both. This confirms that the track can be solved largely by rendered appearance; it does not test reference selection among multiple parts.

Part-type. Each of 67 pairs retains the target and underlying geometry while swapping the labels of the matching candidate and a nonmatching candidate. The option list is fixed within the pair. Geometry pixels match exactly, and pixel changes lie within the label masks. Two independently regenerated canaries are byte-identical. Source part-type equality derives a candidate gold; whether exactly one match is visually identifiable is checked separately by

humans.

The 67 auxiliary position panels display a target and four candidate panels without text embedded in the image. Options refer to candidate positions. The comparison changes reference interface, separation and visual access together, so it cannot isolate a causal OCR effect. Panels have their own human decisions and do not enter paired Both.

4.2. Assignment and restricted shortcut audits

Candidate construction enumerates four-option permutations and admissible nonmatching swaps with visible anchors and a unique source-type match. For Part-type, 4,824 candidates reduce to 2,412 constraint-equivalent configurations. A finite assignment selects one per parent under pre-committed bounds: each arm has 16 or 17 golds per option position; each off-diagonal A-to-B transition has five or six pairs; diagonal transitions have zero pairs.

Every fixed-choice rule is bounded by .30 equal-source endpoint accuracy. Thirteen declared oracle-A rules are bounded by .40 B accuracy in both micro and equal-source estimates. These privileged audits receive the correct A answer; some also receive A-side source geometry. They inspect option order, label numbers or spatial relations, without reading B pixels. They are construction diagnostics, not legal single-image baselines or upper bounds on models.

The frozen first-alternative rule attains 23/67 Both. Because each transition cell has at most six pairs, any fixed mapping from A choice alone to B choice has in-sample B accuracy at most 24/67. Additional features fall outside that bound. Color was not jointly optimized, and its strongest declared oracle rules reach 32/73. The full transition matrices and rule outputs remain public.

4.3. Exact-input independent human qualification

All 347 visual observations are frozen before qualification. Each receives two judgments from six distinct reviewers. A reviewer sees at most one observation of a parent, including the auxiliary panel; A/B reviewer sets are disjoint. Queues show one exact observation with native pixels, while hiding the source record, gold, condition, pair membership, other reviews and model outputs.

Reviewers provide a visual choice, one of decidable/ambiguous/ unanswerable/contradictory, and checks for unique target reference, unique visible answer, sufficient visibility and absence of reference contradictions. They attest to native-size inspection and independence. Any decision, choice or structured-check disagreement requires an additional independent adjudicator. A final decidable choice in conflict with generated gold returns the observation for revision and renewed qualification; gold is not silently replaced.

The protocol requires a final decision for every planned visual endpoint before collection. All endpoints remain in

the full denominator; an additional qualified subset contains pairs with two finally decidable endpoints. Raw agreement, choice/check agreement, kappa, exclusions and source coverage are reported. At present, no genuine human judgments have been collected, so answerability is an open acceptance requirement. Machine consistency does not satisfy it.

5. Evaluation Metrics

5.1. Correctness, adaptation and validity

For N fixed pairs, let p_{Ai}, p_{Bi} be normalized predictions, $y_{Ai} \neq y_{Bi}$ the gold choices, and v_{Ai}, v_{Bi} valid-format indicators. Invalid or missing output never matches a gold. Define

$$AAcc = \frac{1}{N} \sum_i [p_{Ai} = y_{Ai}], \quad (1)$$

$$NewAcc = \frac{1}{N} \sum_i [p_{Bi} = y_{Bi}], \quad (2)$$

$$Both = \frac{1}{N} \sum_i [p_{Ai} = y_{Ai}][p_{Bi} = y_{Bi}]. \quad (3)$$

Both is the primary pair outcome. AAcc exposes baseline competence; NewAcc exposes performance on the changed observation. Micro conditional adaptation divides the both-correct count by the A-correct count. A zero denominator is undefined, not zero.

Old-gold retention is $N^{-1} \sum_i v_{Bi} [p_{Bi} = y_{Ai}]$. Together, new-gold correctness, old-gold retention, other valid wrong answers and invalid B output partition the B outcomes. Retention is an observed answer pattern, not evidence of an internal memory mechanism. We report valid A, valid B and valid-both rates alongside semantic metrics. Stable invalid strings do not count as semantic consistency.

The supplemental Part-type example has GT token pair (C, D) . Only matching both tokens earns Both=1. Returning the A gold on B instead contributes to old-gold retention. These statements define the scorer; no displayed GT is a claimed model response.

5.2. Aggregation and uncertainty

For outcome z_{si} from pair i of source s , the primary estimator is

$$\hat{\mu}_{source} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{n_s} \sum_{i=1}^{n_s} z_{si}. \quad (4)$$

Micro rates and eligible pair/source counts accompany it. Conditional source rates are computed within sources with A-correct pairs, then averaged; they are not the ratio of two source-macro scores. Descriptive 95% intervals use 10,000 whole-source bootstrap draws with seed 20260920. Paired model differences use the same items.

Sources share authors and common parts. Sensitivity analysis uses 18 primary-author groups and 17 contributor-connected groups, reporting equal-author means, clustered intervals and leave-one-author-out differences. Full and qualified-subset results are reported together; their difference uses shared source resampling and recomputes each mean on its eligible sources. Empty subsets remain undefined. Exact rational point comparisons retain ties in AAcc and Both ranks. Rank reversals and unadjusted pairwise intervals are exploratory. Intervals crossing zero do not support a directional ranking.

5.3. Requests, failures and repeat policy

Every scheduled request retains its input hash, pre-call intent, provider receipt or explicit failure, normalized answer and inclusion record. Refusals, timeouts, API failures, malformed outputs and missing receipts remain incorrect in the planned denominator. Absent calls cannot disappear through filtering. Repeated invocations, if collected, must all be retained under a frozen policy; source intervals alone do not measure cross-call variation. The existing evaluator implements a sealed three-model, one-attempt study. The broader proposal below requires a new version with its own adapters and complete run manifest.

6. Planned Experiments

6.1. Research questions and model coverage

We organize the study around three questions: (i) does Both reveal differences not apparent in AAcc; (ii) how do image withdrawal, qualification and reference interface affect those differences; and (iii) do structural controls distinguish behavior that natural graph accuracy obscures? No model outcome is assumed by these questions.

Table 2 proposes 15 distinct models: seven proprietary VLMs, six open-weight VLMs and two open-weight text-only models. The inventory follows the repository’s September 20, 2026 planning snapshot; it is not a claim of current endpoint availability. Model revision, provider, precision, image processing and collection window must be checked and committed before execution. A hosted open-weight endpoint is a specific service configuration, not automatically equivalent to its original weights.

Access is planned through a mixture of hosted APIs and self-hosting. The budget snapshot lists 12 models on Open-Router and three for self-hosting; listing alone does not verify live account access, image delivery or backend revision stability. We compare advertised model classes without assigning a universal “reasoning/nonreasoning” label to proprietary systems. Only supported, explicitly frozen thinking or effort modes enter the mode comparison.

6.2. Primary visual results and error decomposition

Table 3 reserves the primary result matrix. Each cell will report an equal-source percentage, with micro rates, counts and clustered intervals in accompanying records. All models receive the same frozen observations and options. Color and Part-type remain separate, including in model comparisons.

Table 4 separates successful adaptation from old-gold retention and invalid output. Conditional adaptation must be read with AAcc and eligible counts. Otherwise a model correct on very few A endpoints could appear strong after conditioning. The full B partition, including other-wrong and invalid, will accompany the compact table. Differences attributable primarily to format or API failures limit a semantic interpretation.

6.3. Image dependence, qualification and interface controls

The no-image condition removes only the image content from each visual request. The 140 stripped A/B pairs have identical messages and different golds. An identical deterministic prediction therefore has Both=0 and symmetric endpoint accuracy at most one half. Independent stochastic calls can produce different answers, so this identity does not force every real no-image run to score zero. The condition tests image dependence, not specific reference selection.

Table 5 records no-image Both, separate panel accuracy and qualified-minus-full changes. Image-minus-no-image paired differences are derived alongside it. A change after qualification must be read with exclusions and source coverage. The panel comparison is descriptive because layout and visual access also change. The current Part-type track has no separately qualified visual answer-preserving intervention. It supports candidate-reference reassignment claims; broader robustness to irrelevant visual changes would require a new control and renewed qualification.

6.4. Text-graph reasoning and algorithmic controls

The supporting task asks for the number of connected components after deleting a named node from an explicit undirected graph, counting isolated survivors. The natural inventory has 73 questions. The hypothetical set contains 73 anchors, 73 answer-changing partners and 73 isomorphic answer-preserving partners.

Twelve strong change pairs across eight sources match surviving nodes, edge count, target degree, complete degree multiset and isolate count. They compare a six-cycle with two triangles, plus identical isolates. The other 61 pairs across 23 sources are degree-visible controls. Their invariance partners are reported separately. These hypothetical edge lists are not newly inferred physical connections. The supplement shows a complete strong example and its independently recomputed component-count GT. CX1–CX2

Table 2. Proposed model inventory. Names and classes are planning metadata; no row represents a completed run. VLMs receive visual, no-image and graph conditions; text-only models receive explicit graphs. The two effort contrasts in Table 7 are configurations, not additional models.

Model	Model class	Planned inputs
GPT-6 Astra	Proprietary VLM	Visual / no-image / graph
GPT-5.6 Sol	Proprietary VLM	Visual / no-image / graph
GPT-5.6 Luna	Proprietary VLM	Visual / no-image / graph
Claude Fable 5.1	Proprietary VLM	Visual / no-image / graph
Claude Sonnet 5	Proprietary VLM	Visual / no-image / graph
Gemini 3.1 Pro Preview	Proprietary VLM	Visual / no-image / graph
Gemini 3.8 Flash	Proprietary VLM	Visual / no-image / graph
Qwen3.5-9B	Open-weight VLM	Visual / no-image / graph
Qwen3.5-27B	Open-weight VLM	Visual / no-image / graph
InternVL3.5-8B	Open-weight VLM	Visual / no-image / graph
InternVL3.5-38B	Open-weight VLM	Visual / no-image / graph
Gemma 3 4B IT	Open-weight VLM	Visual / no-image / graph
Gemma 3 27B IT	Open-weight VLM	Visual / no-image / graph
Qwen3-32B	Open-weight text LM	Graph
DeepSeek-R1-Distill-Qwen-32B	Open-weight text LM	Graph

Table 3. Planned primary visual results (% , equal-source). A and New are endpoint accuracy; Both requires both endpoints correct. All result cells are intentionally empty. Companion records will include micro scores, denominators and source intervals.

Model	Color A	Color New	Color Both	Part-type A	Part-type New	Part-type Both
GPT-6 Astra						
GPT-5.6 Sol						
GPT-5.6 Luna						
Claude Fable 5.1						
Claude Sonnet 5						
Gemini 3.1 Pro Preview						
Gemini 3.8 Flash						
Qwen3.5-9B						
Qwen3.5-27B						
InternVL3.5-8B						
InternVL3.5-38B						
Gemma 3 4B IT						
Gemma 3 27B IT						

instead use a larger induced source graph and require optimization or a conditional policy; those examples remain outside the planned graph result matrix.

Deterministic audits already establish the relevance of this partition. An isolates-plus-one heuristic solves 72/73 natural questions and 61/61 degree-visible change pairs, but 0/12 strong change pairs. BFS and union-find solve all. Anonymous uniform-feature 1-WL cannot distinguish the two strong endpoints; a predictor restricted to that signature must return the same answer to different golds. This restriction excludes unique-ID, positional and higher-order features and does not describe a trained language model’s internals. The supplement reports the full algorithm audit.

Table 6 reserves learned-model results for all four parti-

tions. Both and symmetric endpoint accuracy are computed within each partition; the compact table shows Both. A repeated wrong answer on invariance is not success. Shared anchors prohibit counting these partitions as independent datasets. The strong subset is a small construction challenge, not a broad graph-reasoning benchmark.

6.5. Reasoning modes and staged collection

Table 7 reserves two within-model contrasts: Qwen3.5-27B thinking off/on and Gemini Pro lower/higher effort. The comparison must retain exact supported settings and include billed hidden reasoning in token and cost accounting. Its differences are computed on identical items with all scheduled outputs retained. The stated modes remain proposals until

Table 4. Planned response diagnostics (% , equal-source). Adapt.: B correctness conditional on A correctness, with eligible counts reported separately. Old: old-gold retention. Valid: both endpoints have valid format. Cells remain empty pending inference.

Model	Color Adapt.	Color Old	Color Valid	Part-type Adapt.	Part-type Old	Part-type Valid
GPT-6 Astra						
GPT-5.6 Sol						
GPT-5.6 Luna						
Claude Fable 5.1						
Claude Sonnet 5						
Gemini 3.1 Pro Preview						
Gemini 3.8 Flash						
Qwen3.5-9B						
Qwen3.5-27B						
InternVL3.5-8B						
InternVL3.5-38B						
Gemma 3 4B IT						
Gemma 3 27B IT						

Table 5. Planned controls. No-image and panel entries are percentages; QA-full entries are percentage-point differences in Both between the both-decidable subset and full set. “Type” denotes Part-type. All measured entries remain empty.

Model	No-image Color Both	No-image Type Both	Panel accuracy	Color QA-full	Type QA-full
GPT-6 Astra					
GPT-5.6 Sol					
GPT-5.6 Luna					
Claude Fable 5.1					
Claude Sonnet 5					
Gemini 3.1 Pro Preview					
Gemini 3.8 Flash					
Qwen3.5-9B					
Qwen3.5-27B					
InternVL3.5-8B					
InternVL3.5-38B					
Gemma 3 4B IT					
Gemma 3 27B IT					

the adapters verify their support.

An initial MVP keeps all 15 models while reducing observations: 12 complete pairs per visual family, their 12 Part-type panels, the corresponding 60 no-image requests, and 54 graph observations. It retains all 12 strong graph parents and six degree-visible parents. One invocation per configuration yields 2,370 requests; the two additional mode configurations raise this to 2,718. Selected IDs, source coverage, subset priors and exact-input qualification must be frozen before any live run. The new provider adapters and subset qualification gate are not yet implemented. This MVP validates delivery, cost and exploratory failure patterns; it does not provide a stable full-dataset ranking.

7. Limitations and Release

BrickAtlas covers two atomic visual families and a small public source collection. Color does not address natural color constancy, and candidate-label reassignment does not establish general referring expression comprehension. CAD equality does not guarantee visible uniqueness. Physical buildability, continuous insertion and dynamics are outside the present evaluation. Public source reuse also leaves contamination and dependence concerns.

The release includes source attribution, exact observation manifests, frozen assignment records, reviewer packages, deterministic audits, scoring code and a 3D inspection interface. Current paper assets and blank experiment tables are generated from an explicit proposal schema; earlier manuscripts and revision material are archived. The ex-

Table 6. Planned text-graph results. The four partition columns report equal-source Both (%); symmetric endpoint accuracy and partition-specific valid-both rates accompany the full records. Format valid reports valid predictions over 219 distinct observations. No model results have been entered.

Model	Strong change	Degree-visible change	Strong invariance	Degree-visible invariance	Format valid
GPT-6 Astra					
GPT-5.6 Sol					
GPT-5.6 Luna					
Claude Fable 5.1					
Claude Sonnet 5					
Gemini 3.1 Pro Preview					
Gemini 3.8 Flash					
Qwen3.5-9B					
Qwen3.5-27B					
InternVL3.5-8B					
InternVL3.5-38B					
Gemma 3 4B IT					
Gemma 3 27B IT					
Qwen3-32B					
DeepSeek-R1-Distill-Qwen-32B					

Table 7. Planned mode contrasts. Score differences are percentage points; output tokens and USD summarize each configuration in the contrast. Both configurations must be reported, with their exact settings. Empty cells indicate unmeasured quantities.

Configuration contrast	Δ Color	Δ Type	Δ Strong	Output tokens	USD
Qwen3.5-27B: thinking on minus off					
Gemini 3.1 Pro: higher minus lower effort					

500 isting three-model protocol and the new 15-model proposal
501 are distinct studies. No old response is transferred to a new
502 stimulus.
503 Human qualification and learned-model evaluation re-
504 main unfinished. Useful diagnostic value would require
505 paired differences that survive the declared input controls,
506 qualification analysis and dependence checks. If Both
507 merely reproduces ordinary-accuracy rankings, or effects
508 are explained by failures and visibility, the interpretation
509 must reflect that outcome. The current release establishes an
510 auditable resource and a concrete experiment design, with
511 the empirical comparisons left open.

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